

# Victor J. Huarcaya Azañon

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## Summary

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Optical engineer & Physicist currently leading optical AIV for ESA's Comet Interceptor instrument Comet Camera (CoCa), with 10+ years of experience in precision optics, interferometry, and optomechanical systems. Background includes high-stability interferometric testbeds and engineering model development supporting ESA's *LISA* gravitational wave observatory. Skilled in tolerance-driven optical integration, cleanroom workflows (ISO 5/7), and ECSS-compliant verification planning. Strong at the optics–mechanics–electronics interface, bridging the gap between optical design and hardware realization to translate scientific requirements into traceable, measurable acceptance criteria.

## Professional Experience

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### Optical AIV Engineer

Bern University

Sept. 2024 – Present

Bern, Switzerland

- Lead optical AIV activities for the Comet Camera (CoCa) on ESA's F-class *Comet Interceptor* mission through Phases C/D established and commissioned ISO 5 cleanroom integration facility.
- Performed optical modelling in Zemax of the testbed (Point source, OAP mirror, folding mirrors) to correlate experimental data with theoretical performance (WFE, MTF), ensuring accurate verification for CoCa and future missions (Ramses).
- Authored and released AIT procedures, test scripts, and verification reports under ECSS configuration control; manage non-conformance reports (NCRs) and coordinate rework to closure.
- Drive technical documentation across multi-site AIV teams, ensuring traceability from instrument requirements to acceptance criteria.

### Postdoctoral Researcher

Max Planck Institute for Gravitational Physics (Albert Einstein Institute)

Oct. 2023 – Sept. 2024

Hannover, Germany

- *LISA* (Laser Interferometer Space Antenna) Phase Measurement Subsystem and *LISA* engineering model: optical payload Assembly, Integration and Testing.
- Successfully integrated and tested *LISA* Phase Measurement Subsystem during Phase B, delivering key project milestones for ESA's space-based gravitational wave observatory.

### Research Scientist

Max Planck Institute for Gravitational Physics (Albert Einstein Institute)

May 2017 – Oct. 2023

Hannover, Germany

- Achieved world-record sub-picometer structural stability ( $< 1 \text{ pm}/\sqrt{\text{Hz}}$  below 1 mHz) in compact interferometers, advancing precision for future space missions.
- Developed 5-DOF test mass optical readout system delivering  $10\times$  higher angular precision than commercial autocollimators at low frequencies.
- Designed and validated novel single-element dual-interferometer architecture, reducing system complexity while maintaining sub-picometer performance.
- Operated and maintained ultra-high vacuum systems ( $10^{-7}$  mbar) for precision interferometry experiments.

### Spacecraft Controller

EUMETSAT (for SERCO)

March 2016 – May 2017

Darmstadt, Germany

- Monitored and commanded *Meteosat-7* spacecraft in 24/7 operational environment, maintaining mission performance through end-of-life phase (Phase E/F).
- Executed spacecraft decommissioning procedures as part of mission close-out team.

### Research Assistant

Centre for Quantum Technologies

Sept. 2013 – Jan. 2016

Singapore

- Designed, aligned, and characterized electro-optical and optomechanical systems for quantum optics experiments.
- Implemented FPGA-based PID controllers for laser frequency stabilization, improving lock stability and bandwidth.

## Education and Certifications

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### Ph.D. in Physics (Dr. rer. nat.)

Leibniz Universität Hannover

2023

Hannover, Germany

- **Thesis:** *Advancements in Optical Readout Technologies: Test Mass Sensing and Laser-Frequency Stabilization Techniques for Optical Compact Interferometry*
- Research conducted at Max Planck Institute for Gravitational Physics (Albert Einstein Institute)

### MSc. in Space Science and Technology

Luleå University of Technology

2013

Luleå, Sweden

- **Thesis:** *Inertial Electrostatic Confinement Devices as an Advanced Electric Propulsion System*
- Research conducted at Institute of Space Systems (IRS) - Stuttgart

### MSc. in Space Techniques and Instrumentation

Université Paul Sabatier – Toulouse III

2012

Toulouse, France

### Licenciatura in Physics (240 ECTS credits, equivalent to BSc. + MSc.)

University of Balearic Islands

2010

Palma, Spain

ESA's International School on Space Optics – Certification (ESTEC, Noordwijk), 2025

ESA's Space Optics Instrument Design and Technology Course – Certification (Poltu Quatu, Italy), 2023

Laser Safety Officer Certification – LZH Laser Akademie (Hannover, Germany), 2023

## Skills

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Languages: Spanish (Native) | English (C1 – Advanced) | German (A1 – Basic)

### Technical Skills:

- **Opto-Mechanical Integration:**
  - End-to-end AIT of optical payloads: telescope structures, interferometric benches, focal plane assemblies
  - Precision positioning systems: hexapods, piezo stages, goniometers, kinematic mounts
  - Environmental test support: TVAC, thermal cycling, vibration campaigns with pre/post optical verification
  - Tolerance budgeting: flow-down from instrument WFE/MTF requirements to component-level alignment specs
- **Optical Metrology:**
  - Alignment tools: theodolites, autocollimators, shear-plate interferometers, alignment telescopes
  - Wavefront sensing: Shack–Hartmann sensors, Fizeau/Twyman-Green interferometers (Zygo-type), PSF/MTF analysis
  - 3D coordinate metrology: Spatial Analyzer software for theodolite-based measurements
- **Electro-Optics & Detection:**
  - Photodetector systems: single-element PDs, QPDs, low-noise transimpedance amplifiers
  - Laser systems: frequency stabilization, AOM/EOM modulation, Class 3B/4 safety compliance
  - Signal chain: ADC/DAC integration, noise characterization (shot noise, electronic noise floors)
- **Software & Analysis:**
  - Optical design: Zemax OpticStudio (ray tracing, tolerance analysis, optimization)
  - Data analysis & automation: MATLAB, Python (NumPy, SciPy, instrument control APIs)
  - Documentation: LaTeX, MS Office, Adobe Illustrator for technical schematics
  - Lab control: Serial/GPIB/TCP-IP instrument communication, digital/analog I/O
- **Project Management:**
  - Preparation of AIT plans, test procedures, alignment reports, and configuration logs
  - Experience with ECSS / ESA project workflows
  - Procurement and technical coordination with optical and lab-equipment suppliers

## Publications

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1. **V. Huarcaya** et al. “ $2 \times 10^{-13}$  fractional laser-frequency stability with a 7-cm unequal-arm Mach-Zehnder interferometer”. *Physical Review Applied*, 20, 024078, 2023. doi:10.1103/PhysRevApplied.20.024078
2. **V. Huarcaya** et al. “Single-Element Dual-Interferometer for Precision Inertial Sensing: Sub-Picometer Structural Stability and Performance as a Reference for Laser Frequency Stabilization”. *Sensors*, 23(24):9758, 2023. doi:10.3390/s23249758
3. **V. Huarcaya** et al. “Five degrees of freedom test mass readout via optical levers”. *Classical and Quantum Gravity*, 37(2):025004, 2019. doi:10.1088/1361-6382/ab5c73
4. Y. Yang, K. Yamamoto, **V. Huarcaya** et al. “Single Element Dual Interferometer for precision inertial sensing”. *Sensors*, 20(17):4986, 2020. doi:10.3390/s20174986
5. S. Aria, [...], **V. Huarcaya** et al. “GLINT – Gravitational-wave laser INterferometry triangle”. *Experimental Astronomy*, 44, 181–208, 2017. doi:10.1007/s10686-017-9558-x
6. **V. Huarcaya**. “Advancements in optical readout technologies: Test mass sensing and laser-frequency stabilization techniques for optical compact interferometry”. *PhD Thesis*, Leibniz Universität Hannover, 2024. doi:10.15488/16938

## Selected Presentations

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**International Conference on Space Optics - ICSO** – “Laser-frequency Stabilization via Locking to a Single-Element Mach-Zehnder Interferometer” (Antibes Juan-les-Pins, France), 2024

**International Conference on Space Optics - ICSO** – “ $2 \times 10^{-13}$  fractional laser frequency stability with a 7 cm unequal-arm Mach-Zehnder interferometer” (Antibes Juan-les-Pins, France), 2024

**13th LISA Symposium** – “Five degrees of freedom Test Mass readout via optical levers” (Online), 2020

**12th LISA Symposium** – “Testing an Optical Readout scheme for future Gravitational Reference Sensors” (Chicago), 2018

**Invited Talk** – “Demonstration of multi-degree of freedom Test Mass readout” (Wuhan, China), 2018

**Alpbach Summer School** – “Quantum Physics and Fundamental Physics in Space” (ESA/FFG, Austria), 2015

## Scholarships

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**Graduate Research Scholarship** – National University of Singapore, 2014

**Erasmus Mundus Mobility Grant** – European Commission, 2012

**Sa Nostra Foundation Grant** – For MSc. studies abroad, Spain, 2010–2011

**SENECA Scholarship** – Spanish Ministry of Education, 2008

## Early Career Highlight

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**ESA Student Parabolic Flight Campaign** – Flight Team Leader for experiment “*Visualization of the Richtmyer-Meshkov instability in zero-g*” (Bordeaux, France), 2004